

WASP-43b

R-2723

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_120418 · created 2026-09-01T23:19:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456035.66362 +/- 0.00028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.012
Transit depth (Rp/Rs)^2	1.94 +/- 0.33 [%]
Orbital Inclination (inc)	81.71 +/- 0.69
Airmass coefficient 1 (a1)	1.1464 +/- 0.0082
Airmass coefficient 2 (a2)	-0.098 +/- 0.037
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	None
Optimal Aperture	3.86
Optimal Annulus	4.96
Transit Duration (day)	0.0484 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.012 vs 0.1594 (deviation 1.05σ)
Duration fit vs published	0.0484 d vs 0.0512 d (deviation 0.7σ)
Mid-time O–C	0.8 min
Depth SNR	7.1
Residual scatter	0.71 %

Light curve

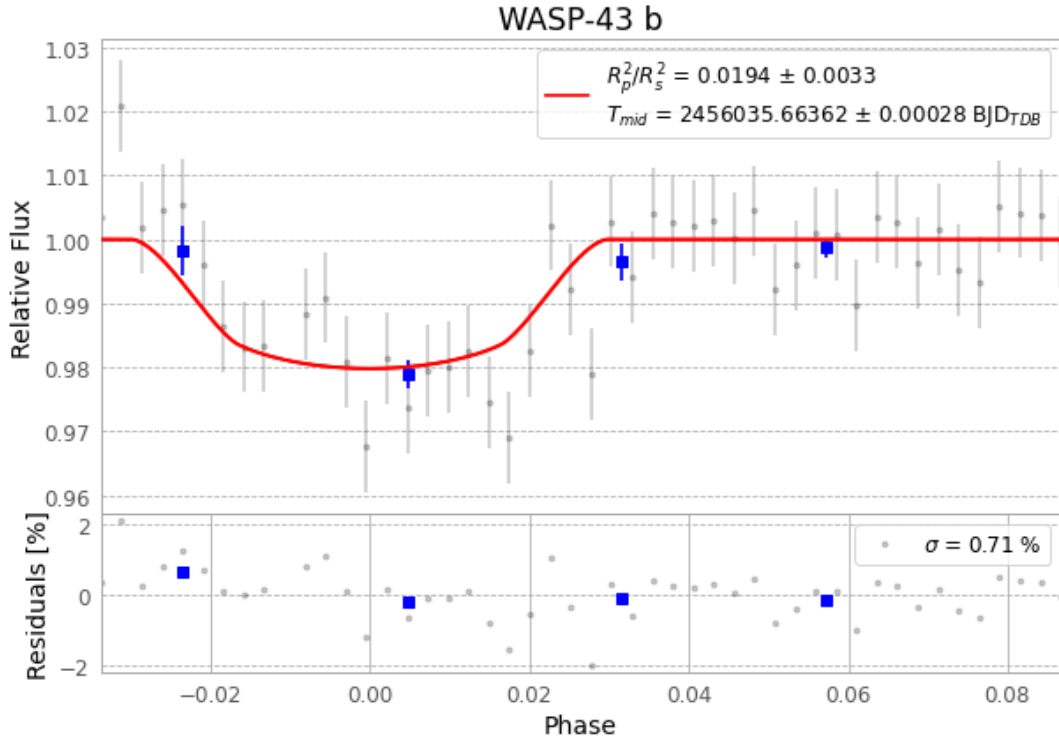


Figure 1 — FinalLightCurve_WASP-43 b_2012-04-17.png (SHA-256 f130631e7629e164...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 283], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3b1b50d40953ea89...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

47 FITS frames (31 MB), WASP-43120418030913.FITS → WASP-43120418053013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-120418.log (b4e54508e94d...), FinalLightCurve_WASP-43 b_2012-04-17.png (f130631e7629...), FinalLightCurve_WASP-43 b_2012-04-17.csv (5f7fa470f1e0...)

Compute resources

Wall time 187.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 23:19Z.